UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	A vector image becomes pixelated when enlarged.	Vector images scale without pixelation.
A2	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.
A3	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A4	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A5	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A6	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A7	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A8	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A9	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.
A10	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A11	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A12	TRUE	Sampling converts a continuous sound wave into digital values.	Sampling measures the wave at intervals.
A13	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A14	TRUE	USB is a plug-and-play connection standard.	USB devices are recognised automatically.
A15	TRUE	Kerning adjusts the space between individual character pairs.	Kerning changes spacing for a specific pair.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	A	Serif	font with small strokes at character ends
B2	M	Storyboard	visual sequence plan
B3	J	Scanner	input device that digitises documents
B4	I	Keyframe	important frame in an animation
B5	O	Frame rate	frames displayed per second
B6	L	Plotter	output device for large drawings
B7	F	Bitmap	image made of pixels
B8	E	Tweening	generating in-between frames
B9	K	Target audience	intended users of a product
B10	H	Browser	software that displays web pages
B11	B	Sampling	converting sound into digital values
B12	N	Multimedia	integration of text, graphics, audio, video and animation
B13	G	Linear multimedia	presentation without user control of sequence
B14	D	Hypermedia	linked multimedia elements
B15	C	Cel animation	hand-drawn frame-by-frame animation

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	B	Compiler
C2	C	Another resource
C3	A	Improves accuracy
C4	B	Loudness
C5	B	In-between frames
C6	C	A video compression family
C7	A	Automatic device recognition
C8	A	Vector
C9	A	Links to other content
C10	C	Display web pages
C11	C	Musical notes and instructions
C12	C	Drawing each frame by hand
C13	B	Original creative works
C14	B	Replace a colour background
C15	C	Play while downloading
C16	A	Smooth jagged edges
C17	A	A visual sequence plan
C18	B	Along a predefined route
C19	C	Card-based
C20	B	A simulated environment
C21	A	Structure web content
C22	B	Drawing input
C23	C	Assemble multimedia elements
C24	B	PNG
C25	B	Pixels
C26	C	Modelling and rendering
C27	B	Messages with a beginning and end
C28	C	Scanner
C29	B	Text, graphics, audio, video and animation
C30	B	Number of pixels in an image

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PART D - WORKED ANSWERS

QUESTION 1

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 2

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

QUESTION 3

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

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QUESTION 4

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(5 marks)

END OF MARKING SCHEME

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